

Suman Dewan – Data Scientist

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Summary

- Over 14 years of professional experience in Data Science and Machine Learning.
- 9+ years of expertise in Python, Machine Learning, and Data Science.
- Proficient in:
 - Building classification-based models
 - Data cleaning and manipulation
 - Exploratory Data Analysis (EDA)
 - Feature Engineering
 - Clustering for unlabeled data
- Skilled in utilizing statistical computer languages for data insights.
- Knowledgeable in:
 - Snowflake
 - Azure Databricks
 - Git
 - RAG
 - Natural Language Processing (NLP)
 - Tensor flow
 - MLOps
- Strong problem-solving skills with a focus on creating reusable artifacts.
- Experienced in waterfall, agile (scrum), and DevOps methodologies.
- Extensive industry experience in:
 - Television Networks
 - Healthcare
 - E-commerce
 - Banking
 - Financial domains
- Proven ability to work effectively with geographically distributed teams.
- Committed to continuous learning and staying updated with industry trends.

Technical Skills

Languages	Proficient Python (including packages Numpy, Pandas, Seaborn, Matplotlib, Plotly, Sklearn, Geopandas, Shapely, Metrics, Datetime, etc.) Experience Java and SQL
Tech Stack	Azure Databricks, Git, Jupyter Notebook (Anaconda), Tensor flow, Logistic regression, Random forest, XG boost, Clustering, NLP, Data visualization, Predictive model building, Exploratory Data Analysis (EDA), Feature Engineering, Cross Validation, SHAP, LLM, RAG

Education

- Masters of Computer Science and Application (MCA)
 - Panjab University, India, 2011
- Bachelors of Science (B.Sc.)
 - DAV College Kangra, India, 2008

Certifications / Professional Awards:

- Certified in Machine Learning

- **Issuing Institution:** Great Learning, Gurgaon, India
- Insta Awards
 - **Details:** Won awards for my performance in project

Professional Experience:

Experience: 9 months

Company: Cognizant

Period: January 2025 to Current

Projects

Project: Causal Factor Analysis for Anomaly Detection

- **Role:** Data Scientist
- **Client:** Leading US-based Television Network
- **Tools/Tech:** Python, SQL, SHAP, Snowflake, Snowpark, Azure SPCs Pipeline, Random Forest
- **Description:** Developed a machine learning solution to identify key causal factors responsible for anomalies in video viewership and impression counts. Anomalies were defined based on threshold breaches and deviation from a 14-day index benchmark.
- **Responsibilities:**
 - Designed and implemented an anomaly detection pipeline to flag significant drops in views or impressions.
 - Used Random Forest to train models and extract top 5 causal features for each anomaly using SHAP (SHapley Additive exPlanations) values to ensure explainability.
 - Created an enriched output table capturing:
 - Top 5 causal factors
 - Current day values
 - Monthly averages and LCL (Lower Control Limit)
 - Deviation percentage vs. average
 - Used Snowflake and Snowpark for data processing and modeling within the data warehouse environment.
 - Collaborated with engineering teams to deploy the solution using Azure SPCs pipeline for automated execution.
 - Ensured code modularity and version control using Git; aligned output with downstream business dashboards.
- **Impact:**
 - Enabled business teams to proactively identify and investigate anomalies across video performance metrics.
 - Improved anomaly investigation turnaround time and supported better root cause analysis with model-driven insights.

Project: Operator Candidate Shortlisting & Resume Analysis (POC)

- **Role:** Data Scientist
- **Client:** US-based Franchise Restaurant
- **Description:** Built two logistic regression models to automate candidate shortlisting for operator roles using structured application data (initial and long forms) and resume text analysis via LLM.
- **Responsibilities:**
 - Extracted and analyzed data from SQL tables using Python (Jupyter); collaborated with data team to understand schema and ensure data quality.
 - For Model A (initial form), built a logistic regression model using 8 key parameters provided by the client; performed EDA and model evaluation using KS value and decile analysis.

- For Model B (long form), integrated variables from multiple tables, narrowed down features through iterations despite small dataset (~545 records), and delivered a robust logistic regression model.
- Developed a Streamlit UI to upload Excel inputs and display model decisions, probability scores, and top 3 influencing features.
- Integrated LLM-based resume analysis to detect presence of business-relevant keywords in resumes and influence selection/rejection decisions using Open AI model and prompt Engineering.
- Ensured explainability, version control via Git, and prepared for downstream API integration.
- **Impact**
 - Successfully demonstrated automation of candidate filtering based on structured and unstructured data.
 - Delivered as a POC; client expressed high satisfaction with model accuracy, usability, and explainability.

Experience: 14 years

Company: Infosys Limited

Period: August 2011 to Jan 2025

Project: Check and improve the retention rate.

- **Role:** Team Lead – Data Scientist
- **Description:** Analyzed new user acquisition and retention rates, implemented strategies to enhance retention.
- **Responsibilities:**
 - Analyzed data field and curated datasets for analysis.
 - Extracted user data using SQL queries from Big Data sources.
 - Used Power BI for generating monthly user reports and retention rate analysis.
 - Conducted Data Cleaning, Exploratory Data Analysis (EDA), and Feature Engineering.
 - Applied Clustering algorithm to identify user segments.

Project: Store Optimization

- **Role:** Data Scientist
- **Description:** Analyzed impact on sales, store operations, and mapped store locations on a USA map.
- **Responsibilities:**
 - Retrieved shape files and visualized the USA map using Python libraries.
 - Analyzed client-provided data using Python (Geopandas, Shapely) for store location plotting.
 - Managed data conversion and analysis on Azure platform.
 - Provided insights on store distribution and profitability.

Project: Sentiment Analysis for Multilingual Model

- **Role:** Data Scientist
- **Description:** Analyzed the customers reviews available in multiple languages and created model to perform the sentiment analysis.
- **Responsibilities:**
 - Retrieved and analyzed customer reviews in .csv format.
 - Used NLP and Bert model to perform the sentiment analysis.

Project: Predicting Heart Disease

- **Role:** Data Scientist
- **Description:** Predicted heart disease occurrences using patient data to help client making recommendations of heart implantable devices.
- **Responsibilities:**
 - Converted SQL data to .csv for analysis.
 - Conducted Data Cleaning, EDA, and Feature Engineering.
 - Implemented Machine Learning models (Logistic Regression, Random Forest).
 - Evaluated model performance using metrics like Confusion Matrix and Recall.

Project: Anti Screening Tool

- **Role:** Data Science Analyst
- **Description:** Developed an application to screen patients for heart implantable devices.
- **Responsibilities:**
 - Converted SL data to .csv for analysis.
 - Conducted Data Cleaning and analysis.
 - Built and validated predictive models for screening decisions.

Prior Experience: Manual and Automation testing (5 years)

- **Description:** Engaged in manual and automation testing across various applications.
- **Responsibilities:**
 - Worked on complete SDLC starting from requirement gathering, test case creation, test data generation, test execution and reporting.
 - Created automation test scripts using Python and Selenium with Java.
 - Coordinated with client for requirement understanding, application overview, test case walkthrough, etc.
 - Worked on multiple applications, such as credit card statement, customer acquisition tool, leadless pacemaker, MRI, and Anti Screening Tool.